FIFTH ANNUAL ECC
UNDERGRADUATE
MATHEMATICS COMPETITION

April 13, 2002

Time control: Three hours

To the team members: The problems are meant to be fun as well as challenging. If you do not succeed in working all ten of them in the allotted time, you are likely to be in good company. Each problem counts 10 points. Partial credit will be given for significant progress or for significant partial solutions, but a thorough job on a few of them will probably be better than some exploratory work on all.

NO BOOKS, NOTES, CALCULATORS, COMPUTERS OR NON-TEAM-MEMBERS may be consulted.

Each team may submit one solution to each problem. Think of your solution as an essay; a logical argument which makes clear why your answer to the question is correct, or why the assertion whose proof is called for in the problem is true.

PLEASE BEGIN EACH PROBLEM ON A NEW SHEET OF PAPER. Team identification and problem number should be clearly given at the top of each sheet of paper submitted.

1. Least squares parabola.

A parabola of the form $y = ax^2$ is to be determined so as to minimize the sum of the squares of the vertical distances from the points (0,0), (1,1) and (2,1) to the parabola. What is the resulting value of a?

2. Mind reading.

In your magic show, you put your volunteer from the audience (who is allowed to use a calculator) through the following calculation (during the year 2002): "Start with the number of days per week that you like to go out. Multiply this number by two. Add 5. Multiply the result by 50. If you have already had your birthday this year, add 1752. If you haven't, add 1751. Now subtract the four-digit year in which you were born." Then have her tell you the resulting number.

(a) If her result is 234, what is her age and how many times per week does she like to go out?

(b) Explain how to get the person's age and desired number of times out per week from the number resulting from the above calculation in the general case, and make clear why it works. Will this work for everyone?

3. $f(f(x)) = 3x^2$.

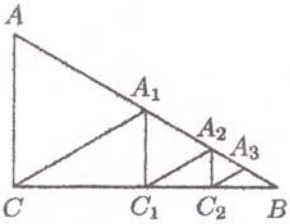
Find a continuous function f from the positive reals to the positive reals such that

$$f(f(x)) = 3x^2.$$

4. The lottery.

Consider a lottery with rules as follows: A computer has randomly chosen four distinct integers of the integers from 1 to 80 inclusive. A player, with no information on what numbers the computer has chosen, chooses eight integers (from 1 to 80), and wins if these eight include all four of the preselected ones. Find the probability of winning, and express it as a rational fraction in lowest terms. Defend your answer.

5. Area ratio.



In triangle ABC, the angle at A is $60°$ and that at C is $90°$. Point A_1 on side AB is chosen so that ACA_1 is an equilateral triangle. Then C_1 is chosen on side BC so that A_1C_1 is perpendicular to BC, and A_2 is chosen on side AB to make $A_1C_1A_2$ an equilateral triangle. This process is continued, resulting in an infinite sequence of equilateral triangles ACA_1, $A_1C_1A_2$, $A_2C_2A_3$, etc. Let U be the area of triangle ACA_1 and V be the sum of the areas of the infinite sequence of equilateral triangles. Find the ratio $\frac{U}{V}$ (and make clear how it is found).

6. Harmonic sums.

Prove that for every positive integer n,

$$1-\frac{1}{2}+\frac{1}{3}-\cdots+\frac{1}{2n-1}-\frac{1}{2n}=\frac{1}{n+1}+\frac{1}{n+2}+\cdots+\frac{1}{2n}.$$

7. Another sum.

Find the sum

$$\frac{1}{1\cdot 4}+\frac{1}{4\cdot 7}+\frac{1}{7\cdot 10}+\cdots+\frac{1}{1999\cdot 2002},$$

and express it in the form $\frac{m}{n}$, where m and n are relatively prime positive integers; i.e., $\frac{m}{n}$ is a quotient of integers in lowest terms.

8. LUB and GLB.

The function F is defined on $(0,\infty)$ by

$$F(x)=\int_0^x \frac{1}{1+t^2}dt+\int_0^{\frac{1}{x}} \frac{1}{1+t^2}dt.$$

Determine whether $F(x)$ is bounded, and if so find its least upper and greatest lower bounds. (These are simply the maximum and minimum values, if these exist.) You must adequately justify your answer.

9. Reciprocal sums.

Given that x, y, and z are nonzero real numbers satisfying

$$\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=\frac{1}{x+y+z},$$

prove that

$$\frac{1}{x^5}+\frac{1}{y^5}+\frac{1}{z^5}=\frac{1}{(x+y+z)^5}.$$

10. A volume.

Find the volume of the region

$$T=\{(x,y,z)\colon |x|+|y|+|x+y|\le 2,\ 0\le z\le x+1\}$$

in Euclidean 3-space.

Solutions, Fifth Annual ECC Undergraduate Mathematics Competition, April 13, 2002

1. Least squares parabola.

The value is $a = 5/17$. The sum of squares described is

$$S(a) = (0-0)^2 + (a-1)^2 + (4a-1)^2,$$

so $S'(a) = 2(a-1) + 8(4a-1) = 34a - 10$, and $S''(a) = 34 > 0$. Thus $S(a)$ is minimized when $34a = 10$; i.e., when $a = 5/17$.

2. Mind reading.

(a) Her age is 34, and she likes to go out two times a week. (Well, another remote possibility is that her age is 134 and she likes to go out once per week.)

(b) Provided that the person is of age less than 100, the resulting number is always a number of three digits or fewer. If we regard it as a three-digit number with possible initial zeros, the first digit is the desired number of times out per week, and the last two digits give the person's age. To see this, let d be the desired number of times out per week and a be the person's age. Assume first that she has already had her birthday this year; thus the year of birth is $2002 - a$. Then the calculation proceeds as follows:

1) d
2) $2d$
3) $2d + 5$
4) $50(2d + 5) = 100d + 250$
5) $100d + 250 + 1752 = 100d + 2002$
6) $100d + 2002 - (2002 - a) = 100d + a$

Assuming that $a < 100$ and $1 \leq d \leq 7$, the result $100d + a$ is a three-digit number with first digit d followed by a written as a two-digit number, with possible first digit 0. If $d = 0$, $100d + a$ may be viewed as a three-digit number with first digit 0.

If she has not yet had her birthday this year, the year of her birth is $2001 - a$, and using 1751 instead of 1752 has the same effect as above. If $a \geq 100$, the first digit is no longer d, but the information would still be recoverable (assuming you have a way of knowing, for example, that the person is 105 rather than 5).

3. $f(f(x)) = 3x^2$.

There are at least two such functions:

$$f(x) = \frac{(3x)^{\sqrt{2}}}{3} \quad \text{and} f(x) = \frac{(3x)^{-\sqrt{2}}}{3}.$$

To find them, try $f(x) = ax^k$. Then

$$3x^2 = f(f(x)) = a(ax^k)^k = a^{k+1}x^{k^2},$$

so we want $k^2 = 2$ and $a^{k+1} = 3$. Then

$$k = \pm\sqrt{2} \quad \text{and } a = 3^{\frac{1}{k+1}} = 3^{\pm\sqrt{2}-1},$$

giving

$$f(x) = 3^{\pm\sqrt{2}-1}x^{\pm\sqrt{2}} = \frac{(3x)^{\pm\sqrt{2}}}{3}.$$

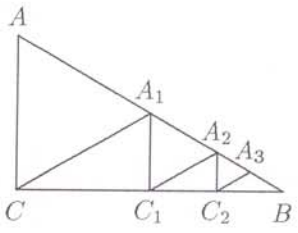
4. The lottery .

The probability is $\frac{1}{22594}$. There are $\binom{80}{8}$ different sets of eight integers which the player may choose. To win, all four of the preselected numbers and any four of the remaining 76 integers must be chosen. There are $\binom{4}{4}\binom{76}{4}$ such selections, so the probability of winning is

$$\frac{\binom{76}{4}}{\binom{80}{8}} = \frac{76!}{4!72!}\frac{8!72!}{80!} = \frac{8\cdot 7\cdot 6\cdot 5}{80\cdot 79\cdot 78\cdot 77} = \frac{1}{79\cdot 26\cdot 11} = \frac{1}{22594}.$$

5. Area ratio.

We show that $\frac{U}{V} = \frac{3}{4}$. From the (30,60,90)-triangle A_1C_1C we see that $A_1C_1 = \frac{1}{2}A_1C = \frac{1}{2}AC$. From this we see that the area of triangle $A_1C_1A_2$ is $\frac{1}{4}$ that of triangle ACA_1, and similarly, the area of each triangle in the sequence is $\frac{1}{4}$ that of its predecessor.



Thus

$$V = U\left(1 + \frac{1}{4} + \frac{1}{4^2} + \cdots\right) = U\left(\frac{1}{1-\frac{1}{4}}\right) = \frac{4}{3}U,$$

and $\frac{U}{V} = \frac{3}{4}$.

6. Harmonic sums.

Call the left- and right-hand sides S_n and T_n, respectively. We see that $S_1 = 1 - 1/2 = 1/2 = T_1$. Suppose that k is an integer for which $S_k = T_k$. Then

$$\begin{aligned} T_{k+1} &= T_k - \frac{1}{k+1} + \frac{1}{2k+1} + \frac{1}{2k+2} \\ &= S_k + \frac{1}{2k+1} - \frac{1}{2k+2} = S_{k+1}, \end{aligned}$$

so by induction the desired result follows.

7. Another sum.

The sum is $\frac{667}{2002}$. Using the fact that $\frac{1}{n} - \frac{1}{n+3} = \frac{3}{n(n+3)}$, we have

$$\begin{aligned} \frac{1}{1\cdot 4} + \frac{1}{4\cdot 7} + \cdots + \frac{1}{1999\cdot 2002} &= \frac{1}{3}\left(1 - \frac{1}{4} + \frac{1}{4} - \frac{1}{7} + \frac{1}{7} - \cdots - \frac{1}{1999} + \frac{1}{1999} - \frac{1}{2002}\right) \\ &= \frac{1}{3}\left(1 - \frac{1}{2002}\right) \\ &= \frac{2001}{6006} = \frac{667}{2002} \end{aligned}$$

8. LUB and GLB.

The function is in fact constant, having constant value $\pi/2$, which is therefore both the greatest lower and least upper bound. We calculate first its derivative, using the chain rule in the second term:

$$\begin{aligned} F'(x) &= \frac{1}{1+x^2} + \left(\frac{1}{1+\frac{1}{x^2}}\right)\left(-\frac{1}{x^2}\right) \\ &= \frac{1}{1+x^2} - \frac{1}{1+x^2} \end{aligned}.$$

Thus, $F'(x) = 0$ for all x, and $F(x)$ is constant. To evaluate the constant we may use

$$F(1) = 2\int_0^1 \frac{1}{1+t^2}dt = 2(\tan^{-1} 1 - \tan^{-1} 0) = \frac{\pi}{2}.$$

Alternately, one may simply evaluate the integrals, to obtain

$$\tan^{-1} x + \tan^{-1}\frac{1}{x} = \tan^{-1} x + \left(\frac{\pi}{2} - \tan^{-1} x\right) = \frac{\pi}{2}.$$

9. Reciprocal sums.

Let's look at the given equation as an equation in z:

$$\frac{1}{z} - \frac{1}{z+(x+y)} = -\left(\frac{1}{x}+\frac{1}{y}\right);$$

$$\frac{x+y}{z(z+x+y)} = -\frac{x+y}{xy}.$$

If $x+y=0$ this holds for all $z \neq 0$. If $x+y \neq 0$ we have $z^2+(x+y)z+xy=0$; i.e., $(z+x)(z+y)=0$. Thus (1) holds if and only if one of $x+y$, $x+z$, $y+z$ is zero. If $x+y=0$, then for all odd n (and in particular for $n=5$)

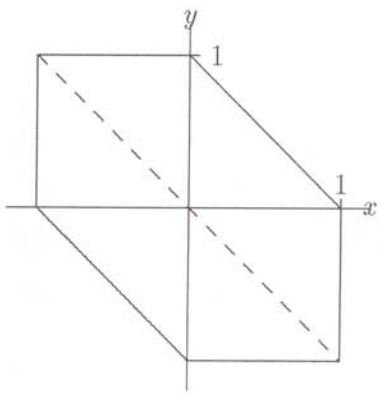
$$\frac{1}{x^n}+\frac{1}{y^n}+\frac{1}{z^n} = \frac{1}{z^n} = \frac{1}{(x+y+z)^n},$$

and similarly if $x+z=0$ or $y+z=0$. ∎

10. A volume.

The volume is 3, as we now show.

Begin by sketching the region in the (x,y)-plane where $|x|+|y|+|x+y| \leq 2$. In the first quadrant this condition reduces to $x+y \leq 1$. In the second quadrant above the line $x+y=0$ the condition is $y \leq 1$ and below the line $x+y=0$ it is $x \geq -1$. This gives us the top half of the region shown at the right, and we may then reflect through the origin to get the bottom half, since the region is symmetric with respect to the origin. Within this region, $x+1 \geq 0$, so the region shown in the (x,y)-plane is the base of T, and the roof of T is the plane $z=x+1$. Thus its volume is given by



$$\begin{aligned} V &= \int_{-1}^{0}\int_{-x-1}^{1}(x+1)dydx + \int_{0}^{1}\int_{-1}^{-x+1}(x+1)dydx \\ &= \int_{-1}^{0}(x+1)(x+2)dx + \int_{0}^{1}(x+1)(-x+2)dx \\ &= \int_{-1}^{0}(x^2+3x+2)dx + \int_{0}^{1}(-x^2+x+2) \\ &= 3. \end{aligned}$$